# Uncomputable finite-size gap exponents*

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Version 1.0.0, October 2026

**Abstract**

We classify the nonnegative finite-size spectral-gap exponents of translation-invariant nearest-neighbour Hamiltonians on the square lattice, with finite local dimension, integer interaction matrices and open boundaries. The gap is measured above the entire ground space on an $L \times L$ box. The existing exponents in $\Delta(L) = L^{-z+o(1)}$ are exactly the nonnegative limit-computable ($\Delta_2^0$) reals. The finite nonnegative lower/upper exponent pairs are exactly the ordered pairs consisting of a left-$\Sigma_2^0$ real and a left-$\Pi_2^0$ real. In particular, one fixed integer-valued model has exponent equal to the halting probability of any chosen prefix-universal machine. A classical boundary preference fixes a size counter and a computation tableau; a reversible mixed-radix counter supplies the small gap. We bound every sector of the full Hilbert space. The closing modes live on a boundary segment of length $O(\log L)$, and the local dimension may depend on the prescribed pair. These are finite-box exponents, with no assertion about bulk dispersion or stability of thermodynamic critical exponents.

## 1 Statement and setting

An asymptotic exponent can be uncomputable even when every finite system is specified by integers. We prove an exact arithmetic classification for the gap on square boxes. The realization uses the freedom of open boundaries: the only dynamics in its low-energy sector takes place on a logarithmic segment of the bottom edge.

Fix a finite local dimension $q$ and three Hermitian matrices $h_0, h_h, h_v$, of sizes $q$, $q^2$, $q^2$, respectively. On $\Lambda_L = \{0, \dots, L-1\}^2$, let

$$H_L = \sum_{x \in \Lambda_L} (h_0)_x + \sum_{x, x+e_1 \in \Lambda_L} (h_h)_{x,x+e_1} + \sum_{x, x+e_2 \in \Lambda_L} (h_v)_{x,x+e_2}. \tag{1}$$

The orientation of each bond is fixed; rotation invariance is not required. There are no additional boundary operators. When the spectrum has at least two distinct eigenvalues, set

$$E_0(L) = \min \operatorname{spec} H_L, \qquad \Delta(L) = \min\{\lambda - E_0(L) : \lambda \in \operatorname{spec} H_L,\ \lambda > E_0(L)\}. \tag{2}$$

We assume this gap is defined for all sufficiently large $L$. Thus scalar finite-volume Hamiltonians are excluded when necessary, and a degenerate ground space does not itself make the gap zero. For $L \geq 2$ put

$$z_L = \frac{-\log \Delta(L)}{\log L}, \qquad z^- = \liminf_{L\to\infty} z_L, \qquad z^+ = \limsup_{L\to\infty} z_L. \tag{3}$$

*The authors developed this work with AI models from more than one developer and take responsibility for every claim. How the work was done, and what the machine checks cover, is described at the end of the paper.

An existing finite exponent is $z = z^- = z^+$.

For a real $a$, the conventions $a \in$ left-$\Sigma_2^0$ and $a \in$ left-$\Pi_2^0$ refer to the arithmetical complexity of the strict lower rational cut $\{r \in \mathbb{Q} : r < a\}$. A real is in $\Delta_2^0$ if it is computable relative to the halting oracle, equivalently if it is a limit of a computable rational sequence. These are representations of real numbers, not conventions about an unspecified digit expansion [13, 6].

**Theorem 1.1** (Exact classification)**.** *Consider* (1) *with integer interaction matrices and arbitrary finite local dimension.*

1. *The existing exponents in* $[0, \infty)$ *are exactly* $\Delta_2^0 \cap [0, \infty)$.
2. *The finite pairs* $(z^-, z^+)$ *in* $[0, \infty)^2$ *are exactly the pairs* $(a, b)$ *with* $0 \le a \le b < \infty$, $a \in$ *left-*$\Sigma_2^0$ *and* $b \in$ *left-*$\Pi_2^0$.

*Each pair is realized by one fixed Hamiltonian. Its local dimension may depend on the pair. The necessity statements hold more generally for algebraic interaction matrices, in any fixed dimension, whenever the gaps and endpoints in question are defined and finite.*

**Corollary 1.2** (One fixed uncomputable exponent)**.** *For any chosen prefix-universal machine* $U$*, there is a fixed model* (1)*, with integer matrices, for which* $\Delta(L) = L^{-\Omega_U + o(1)}$*, where* $\Omega_U$ *is the halting probability of* $U$*. No uncomputable number occurs in a matrix entry.*

The construction and its full spectral estimate are in Sections 3–5. Section 6 provides the required computationally fast words, and Section 7 proves the converse.

### 1.1 Related work and the meaning of the exponent

Cubitt, Pérez-García and Wolf proved undecidability of the spectral-gap problem for translation-invariant two-dimensional systems [3, 4]. Bausch, Cubitt and Watson established uncomputable phase diagrams for continuous one-parameter families [2]. Purcell, Li and Cubitt constructed a phase transition whose location is a Chaitin halting probability [10]. Here the uncomputable number is the finite-box exponent of a single fixed Hamiltonian. A transition location and an exponent are different data.

Our boundary initialization implements the tiling-bonus mechanism of Bausch, Cubitt, Lucia, Pérez-García and Wolf [1], whose translation-invariant interactions favour a particular initialization at an open edge. We give the particular four-flag identity and its proof, rather than attributing that identity verbatim to their construction. Watson and Cubitt study complexity encoded in the ground-state energy density of one fixed Hamiltonian [11]; their quantity and computational question are distinct from the finite-size exponent here.

The arithmetic of densities is prior work: Zheng and Weihrauch develop the real hierarchy [13], and Downey, Jockusch and Schupp classify lower/upper density pairs of computable sets [6, Theorem 5.1 and Corollaries 5.4–5.5]. We supply a fast-word refinement and a Hamiltonian realization. The companion work *The uncomputable price of precision* [5] treats an arithmetic law for games. The proof below is independent of that work.

Xu et al. [12] distinguish the algebraic bulk gap from boundary excitations and give certified upper approximations in their bulk setting. Our closing modes occupy $O(\log L)$ sites of a boundary row. Theorem 1.1 consequently does not classify bulk dispersion, universal critical behavior, or stable thermodynamic critical exponents. It also makes no general exclusion of negative finite-size exponents; necessity still applies to any such finite exponents. We make no exhaustive priority claim.

## 2 Two elementary spectral and boundary estimates

We first record estimates that will control all computational and noncomputational sectors. Graph Laplacians use the quadratic form $\sum_{uv\in E}|\psi_u-\psi_v|^2$, with repeated edges counted with their integer weights.

**Lemma 2.1** (Penalty localization)**.** *Let $C$ be a finite connected graph of maximum degree at most $D$, and let $V$ be diagonal with values in $\{0\}\cup[1,\infty)$. If all vertices are penalized, then $L_C+V\geq I$. If both free and penalized vertices occur, and each connected free cluster has at most $g$ vertices, then*

$$L_C+V \;\geq\; \frac{1}{8Dg^2}I. \tag{4}$$

*A free path on $n\geq 2$ vertices has positive gap $2-2\cos(\pi/n)\geq 4/n^2$; a free cycle on $n\geq 3$ vertices has positive gap $2-2\cos(2\pi/n)\geq 4/n^2$. A free isolated vertex has no positive eigenvalue. The two-vertex cycle with two parallel unit edges has gap 4.*

*Proof.* The all-penalized case is immediate. For each free cluster $K_i$, choose an edge $u_ip_i$ to a penalized vertex. Each penalized vertex is selected at most $D$ times, so half of its penalty can allocate $1/(2D)$ to each selected cluster. The remaining half pays for its own norm. Therefore the energy of $\psi$ is at least

$$\sum_i\left(\sum_{uv\in E(K_i)}|\psi_u-\psi_v|^2+|\psi_{u_i}-\psi_{p_i}|^2+\frac{|\psi_{p_i}|^2}{2D}\right)+\frac{1}{2}\|\psi_{C\setminus F}\|^2,$$

where $F$ is the free set. Let $M_i=\max_{v\in K_i}|\psi_v|$. If $|\psi_{p_i}|\geq M_i/2$, the allocated penalty is at least $M_i^2/(8D)$. Otherwise a path from a maximum to $p_i$, of at most $g$ edges, has energy at least $M_i^2/(4g)$ by Cauchy–Schwarz. Since $\|\psi_{K_i}\|^2\leq gM_i^2$, either case bounds the cluster norm by (4). The path and cycle statements follow from their explicit spectra. Keeping parallel-edge weights gives the last case; larger integer weights cannot reduce the lower bounds. □

**Lemma 2.2** (Boundary preference)**.** *At each site put four flag bits $T,F,B,R$. Give a site energy $-(T+F+B+R)$. On a vertical bond, with upper endpoint $x$ and lower endpoint $y$, add $2T_y+2B_x$; on a horizontal bond with left endpoint $x$ and right endpoint $y$, add $2F_y+2R_x$. Add unit penalties for forbidden onsite and bond patterns. Then*

$$E_{\mathrm{cl}}(\sigma)+4L=\#\{\textit{incorrect flag bits}\}+\#\{\textit{violated local rules}\}. \tag{5}$$

*Here a correct flag is the indicator of its corresponding physical edge. If exactly one configuration has correct flags and no rule violation, it is the unique classical minimizer, of energy $-4L$, and every other configuration has energy at least $-4L+1$.*

*Proof.* The coefficient of $T$ is $-1$ on the top row and $+1$ elsewhere, because every other site has one bond from above. Its energy is therefore $-L$ plus the number of errors relative to the top-row indicator. Apply the same argument separately to $F,B,R$ and add the rule penalties. □

All bond contributions in this lemma are genuine two-site matrices, even when one factor is the identity. The edge preference comes solely from missing bonds in (1). We have proved a classical gap *at least* one.

# 3 A unique classical tableau from uniform interactions

Fix an integer $\beta \geq 1$ and a computable binary word $A(0), A(1), \ldots$ whose bits can be written sufficiently quickly; Section 6 constructs every word we need. Fix a total deterministic one-tape machine $M$ that writes the word on a persistent output track, with bit $j$ at column $j$, and never crosses the left edge. It can be arranged that the first $j$ bits are written by time $2^{j-1} - 2$ for every sufficiently large $j$. This is a timing upper bound, not a time supplied as a coupling.

Besides the four flags, the classical alphabet contains a binary digit $b$, a carry $c$, a past-one flag $f$, and the tape, head, persistent output and arrival fields of $M$. Every listed rule is enforced by a diagonal onsite or nearest-neighbour unit penalty. Conjunctions can be imposed as a single forbidden-pattern projector; their precise grouping does not affect (5).

## 3.1 Counting the row number

The counter rules are

$$\begin{aligned} &T = 1 \implies b = f = 0, && F = 1 \implies c = 1, \\ &F_y = 0 \implies c_y = b_x c_x && \text{on a horizontal bond } x \to y, \\ &T_y = 0 \implies b_y = b_x \operatorname{XOR} c_x, \quad f_y = f_x \operatorname{OR} b_x && \text{on a vertical bond } x \to y. \end{aligned}$$

With correct flags, carries are fixed left to right on each row, and the vertical rule advances the binary row number. Induction gives row $r$ equal to the binary expansion of $r$, with least significant bit at column zero; $f$ records whether that column was one on an earlier row. Thus the derived label

$$\mathsf{reg} = B\,(f \operatorname{OR} b) \tag{6}$$

selects exactly columns $0, \ldots, \ell(L) - 1$ on the bottom row, where

$$\ell(L) = \operatorname{bitlength}(L - 1) = \log_2 L + O(1). \tag{7}$$

Indeed a column $j$ has shown a one by row $L - 1$ exactly when $2^j \leq L - 1$. For $L \geq 2$, $\ell(L) < L$.

## 3.2 A nearest-neighbour computation tableau

Let the tape field be $\gamma$, the head state be $h \in Q \cup \{\perp\}$, the output be $o \in \{\perp, 0, 1\}$, and the two arrival signals be $a_L, a_R \in Q \cup \{\perp\}$. The top row has blank tape and output, with the initial head only at its leftmost site. On each horizontal bond, the left site's transition fixes the right site's $a_L$; the right site's transition fixes the left site's $a_R$. A missing source gives a blank arrival. Require $a_L = \perp$ at $F = 1$ and $a_R = \perp$ at $R = 1$. These equalities fix every unused arrival signal as well.

On a vertical bond, the upper site's own transition fixes the tape and output below it. Its staying-head transition, or its $a_L$ or $a_R$, fixes the next head; two incoming heads are forbidden. This is a finite rule table determined by the transition function of $M$. In particular, the arrival fields implement the update by horizontal and vertical *two-site* tests; a three-site update is not an interaction.

Given one row's tape, head and output, first fix all arrival fields by the horizontal tests and edge initialization, then fix the next row by the vertical tests. This proves uniqueness by induction, including the last row's arrival fields. At time $t$ the head column is at most $t$. A transition to another included row cannot leave the right side of an $L \times L$ tableau. An outgoing move after the final row places no condition on a nonexistent next row. Consequently the tableau exists and is unique for every $L \geq 2$, regardless of whether all desired output bits have yet appeared.

Together with the counter, this gives a unique correct classical configuration $\sigma_L^*$ and the unit separation of Lemma 2.2. On the bottom register put

$$m_j = \beta \mathbf{1}_{\{o_j=1\}}, \qquad r_j = 2^{m_j}, \qquad 0 \le j < \ell(L). \tag{8}$$

For all sufficiently large $L$, the timing assumption gives $o_j = A(j)$ for $j < \ell(L)$: the first $\ell$ bits appear by row $2^{\ell-1} - 2$, and $L - 1 \ge 2^{\ell-1}$. Therefore

$$k(L) = \sum_{j<\ell(L)} m_j = \beta \, |A \cap [0, \ell(L))|. \tag{9}$$

# 4 The quantum counter and all its sectors

At every site use the same additional space with digit $d \in \{0, \dots, 2^\beta - 1\}$ and head-side symbol $s \in \{\mathsf{NL}, \mathsf{NR}, \mathsf{INC}, \mathsf{RET}\}$. The symbols $\mathsf{NL}$ and $\mathsf{NR}$ mean left and right of a head; the other two are head states. Outside $\mathsf{reg}$, penalize every dynamic field other than $(0, \mathsf{NL})$ by one and apply no transition. Inside $\mathsf{reg}$, penalize digits $d \ge r_j$ and, on two consecutive register sites, the side-symbol patterns

$$(\mathsf{NR}, \mathsf{NL}), \qquad (\text{head}, \mathsf{NL}), \qquad (\mathsf{NR}, \text{head}), \qquad (\text{head}, \text{head}).$$

Either head state is meant here. All these penalties are nonnegative.

For each transition $u \leftrightarrow v$ below, add its edge Laplacian $(|u\rangle - |v\rangle)(\langle u| - \langle v|)$. Digits not explicitly changed are retained. The rules on the intended register are

$$(d, \mathsf{INC}) \leftrightarrow (d+1, \mathsf{RET}) \qquad (d+1 < r_j), \tag{10}$$

$$(r_j - 1, \mathsf{INC})(e, \mathsf{NR}) \leftrightarrow (0, \mathsf{NL})(e, \mathsf{INC}) \qquad (j+1 < \ell), \tag{11}$$

$$(r_j - 1, \mathsf{INC}) \leftrightarrow (0, \mathsf{RET}) \qquad (j = \ell - 1), \tag{12}$$

$$(0, \mathsf{NL})(e, \mathsf{RET}) \leftrightarrow (0, \mathsf{RET})(e, \mathsf{NR}) \qquad (j+1 < \ell), \tag{13}$$

$$(d, \mathsf{RET}) \leftrightarrow (d, \mathsf{INC}) \qquad (j = 0). \tag{14}$$

The conditions involving $j$ in this notation are implemented as follows. Rules (10) and (14) are onsite, guarded by $\mathsf{reg} = 1$, with the latter also requiring $F = 1$. Rules (11) and (13) require both endpoints to be register sites. Rule (12) is a horizontal bond term: its left endpoint has $\mathsf{reg} = 1$ and its right endpoint has $\mathsf{reg} = 0$; it changes only the left dynamic field and is the identity on the right dynamic field. Such a right neighbour exists on $\sigma_L^*$ since $\ell(L) < L$. Every digit test uses finite local classical labels. Thus every rule is onsite or nearest-neighbour, including the last-cell test. No rule queries coordinates or the absence of an unrepresented neighbour.

Write $K_L$ for this nonnegative dynamic operator on the correct classical block. In the next lemma set $B_* = \prod_{j<\ell} r_j = 2^k$; this symbol is unrelated to the bottom flag $B$.

**Lemma 4.1** (Full counter gap)**.** *For $\ell \ge 1$ the dynamic kernel is nonempty, and the smallest positive eigenvalue obeys*

$$\frac{1}{64\ell^2 B_*^2} \;\le\; \Delta^+(K_L) \;\le\; \frac{4\pi^2}{B_*^2}. \tag{15}$$

*There is a free counter orbit of length*

$$N = 2\sum_{j=0}^{\ell-1} \frac{B_*}{r_0 \cdots r_{j-1}}, \qquad 2B_* \le N \le 2\ell B_*. \tag{16}$$

*The empty product is one. If $N = 2$, the orbit retains two parallel edges.*

*Proof.* First fix the exterior fields. A wrong exterior field cannot change under any transition and carries a penalty at least one, so the entire corresponding block is at least one. With the exterior correct, head number on the register is conserved.

With zero heads there are no transitions. Free isolated configurations, including the all-zero, all-NL configuration, give ground states; penalized isolated configurations cost at least one. With two or more heads every configuration is penalized: between consecutive heads the side-symbol rules would require NR immediately after the first and NL immediately before the second, forcing a forbidden switch. Adjacent heads are forbidden directly. The all-penalized part of Lemma 2.1 applies even if this sector has large degree.

With one head, there is at most one forward match and one backward match. At an incrementing head the forward choice is increment, carry or wrap; its predecessor is a carry or restart. At a returning head, its digit distinguishes an increment predecessor (nonzero) from a wrap or return predecessor (zero, at or away from the last site). The return or restart successor is also unique. Illegal digits or side strings remove possible matches. Hence transition degree is at most two, counting weights.

There are exactly $2\ell B_*$ free one-head configurations: choose a head position, one of its two states, and legal digits; the side symbols are forced. On a component meeting penalties, a free cluster has size at most $2\ell B_*$, so Lemma 2.1, with $D = 2$, gives $1/[16(2\ell B_*)^2]$. On a completely free component, a nontrivial path or cycle has at most $2\ell B_*$ vertices and the positive spectral gap is larger than the same bound. Free singletons only enlarge the ground space. These arguments prove the lower bound in every sector.

For the upper bound start with an incrementing head at site zero and any legal digit string. It increments the mixed-radix value, carries over maximal digits, and returns over the zeros it has created. Overflow wraps to zero. It returns to the initial digit value after $B_*$ increments. Each cell $j$ is visited in the forward sweep $B_*/(r_0 \cdots r_{j-1})$ times and contributes twice that many transitions in a complete orbit. This gives (16). The orbit is free and its positive gap is $2 - 2\cos(2\pi/N) \le 4\pi^2/B_*^2$. When $\ell = 1, r_0 = 1$, wrap and restart contribute to the same pair of states, giving the double-edge cycle with gap four. This case satisfies the same inequalities. □

We use the orbit only for an upper bound. Equality of the full gap with its cycle eigenvalue is unnecessary and is not asserted. In particular, other free one-head components and their ground states are allowed.

## 5 Assembly and the realization formula

The complete local space is the tensor product of the finite classical alphabet and the dynamic alphabet. Add the classical terms to the guarded dynamic terms. All matrix entries are integers: diagonal rules are integer projectors and each transition contributes diagonal $+1$ and off-diagonal $-1$. The matrices are fixed once $M$ and $\beta$ are chosen. They have exactly the form (1).

**Theorem 5.1** (Boundary realization)**.** *For every word satisfying the timing condition of Section 3 and every integer $\beta \ge 1$, the constructed Hamiltonian has, for all sufficiently large $L$,*

$$\frac{4^{-k(L)}}{64\ell(L)^2} \;\le\; \Delta(L) \;\le\; \min\{1, 4\pi^2\, 4^{-k(L)}\}. \tag{17}$$

*Consequently, writing $\rho_J(A) = |A \cap [0, J)|/J$,*

$$z_L = 2\beta\rho_{\ell(L)}(A) + O\Big(\frac{\log\log L}{\log L}\Big). \tag{18}$$

*In particular* $z^- = 2\beta \liminf_J \rho_J(A)$ *and* $z^+ = 2\beta \limsup_J \rho_J(A)$.

*Proof.* All classical labels are conserved. The exact block decomposition is

$$H_L = \bigoplus_\sigma (E_{\mathrm{cl}}(\sigma) I + K_\sigma), \qquad K_\sigma \geq 0. \tag{19}$$

The correct block has energy $-4L$ and a nonempty kernel. Every other classical block is at least $-4L + 1$, independently of its apparent register, head pattern or output. A superposition across blocks cannot evade this operator inequality. Within the correct block exterior defects, zero-head excitations, multi-head sectors and all one-head components were covered in Lemma 4.1. Thus

$$\min\{1, \Delta^+(K_L)\} \leq \Delta(L) \leq \Delta^+(K_L).$$

The lower estimate in (15) is at most one and proves the lower bound in (17). Changing a single frozen exterior field from $(0, \mathsf{NL})$ to $(0, \mathsf{NR})$ in a dynamic ground state produces a unit-energy eigenvector. There are no register/exterior adjacency penalties and the wrap term is the identity on the exterior dynamic field, so this gives $\Delta(L) \leq 1$. The orbit gives the other upper bound.

Taking base-two logarithms of the bracket gives $\log_2(1/\Delta(L)) = 2k(L) + O(\log \ell(L))$. Equations (7) and (9) yield (18). Every sufficiently large integer occurs as $\ell(L)$, and $\ell(L)$ increases only by unit steps, which proves the endpoint equalities. □

Wrong tableaux, fake interior boundaries and arbitrarily long apparent computations are covered by the classical unit separation, without any runtime bound on those wrong blocks. The only true computation is static. The closing eigenvectors keep that tableau fixed, freeze the exterior, and evolve the quantum counter on the logarithmic boundary segment.

## 6 Fast words with prescribed density endpoints

The usual computable-density classification supplies the endpoint arithmetic [6]. The following timing refinement is needed because only $L$ rows are available to write approximately $\log_2 L$ output bits.

**Lemma 6.1** (Fast density word)**.** *If* $0 \leq a \leq b \leq 1$, $a \in$ *left-*$\Sigma^0_2$ *and* $b \in$ *left-*$\Pi^0_2$*, there is a binary word* $A$ *with lower density* $a$ *and upper density* $b$ *whose* $j$*th bit is computable in time polynomial in* $\log(j + 2)$*. The polynomial and its constants may depend on the endpoints' programs.*

*Proof.* The sequence characterizations in [6, Theorem 5.1 and Corollaries 5.4–5.5] give computable rational sequences with liminf $a$ and limsup $b$. If $a < b$, choose a fixed rational $c$ with $a < c < b$. Clip the first sequence to $[0, c]$ and the second to $[c, 1]$, and interleave them. Continuity and monotonicity of clipping preserve the desired endpoint of each sequence. The result is a computable rational sequence $a_n \in [0, 1]$ with endpoints $a, b$. If $a = b$, membership of both cuts gives a $\Delta^0_2$ real, so use a convergent computable rational sequence clipped to $[0, 1]$. Choosing the separator uniformly from arbitrary endpoint names is not claimed.

Fix $a_0 = 0$. Put $\sigma(0) = 0$. At stage $s$, simulate the canonical algorithm for $a_{\sigma(s)+1}$ from its start for $s$ steps, and advance $\sigma$ by one exactly when the output has appeared. Otherwise retain its value. Totality makes $\sigma$ unbounded, and unit advances preserve liminf and limsup for $b_s = a_{\sigma(s)}$. Every selected output other than $a_0$ has $O(s)$ bits and comes from a bounded simulation. Reconstructing all advances to stage $s$ takes $O(s^2)$ simulated steps, with polynomial universal-simulation overhead. This explicit output bound prevents a large rational name from defeating the slowdown.

Let $N_0 = 1$ and $N_{s+1} = (s+2)N_s$. Set $A(0) = 0$ and, for $0 \le t < N_{s+1} - N_s$, put

$$A(N_s + t) = \lfloor (t+1)b_s \rfloor - \lfloor tb_s \rfloor. \tag{20}$$

The number of ones in the first $t$ letters of that stage is $\lfloor tb_s \rfloor$. If $J = N_s + t$, its density differs by less than $1/J$ from

$$\frac{N_s}{J}\rho_{N_s}(A) + \frac{t}{J}b_s. \tag{21}$$

At the end of the stage, the density differs from $b_s$ by at most $1/(s+2) + 1/N_{s+1}$. Endpoints therefore attain the liminf and limsup of $b_s$. The convex formula shows that intermediate densities cannot create additional limit extrema. This proves both density assertions, including $a = 0$ or $b = 1$.

To compute $A(j)$, find its factorial-sized stage, reconstruct $\sigma(s)$ and its bounded rational output, then evaluate the two floor values in (20). The stage has $s = O(\log(j+2))$, all index arithmetic has $O(\log(j+2))$ bits, and the selected rational has polynomially bounded length. All steps take polynomial time in $\log(j+2)$. □

*Realization in Theorem 1.1.* Given $(z^-, z^+) = (a, b)$ as in the theorem, choose an integer $\beta \ge \max\{1, b/2\}$. Positive rational scaling preserves the cut classes. Apply Lemma 6.1 to $(a/(2\beta), b/(2\beta))$. A one-tape machine can compute the bits and write them successively on a persistent track, using separate work symbols. Simulation and travel to column $j$ cost at most polynomial time in $j$ per bit. Its prefix runtime is therefore eventually less than $2^{j-1} - 2$. Theorem 5.1 gives the prescribed pair. Taking $a = b$ gives every existing nonnegative $\Delta_2^0$ exponent. □

*Proof of Corollary 1.2.* Dovetail the prefix-universal machine's programs and enumerate its halting weights, giving computable increasing rational sums $\Omega_{U,s}$ converging to $\Omega_U$. Use the convergent sequence $\Omega_{U,s}/2$ in the word construction and set $\beta = 1$. The resulting exponent is $2(\Omega_U/2) = \Omega_U$. The finite simulator and word algorithm determine the finite classical alphabet and integer matrices. The real is encoded in an asymptotic density, not as a local coefficient. □

An exponent zero does not assert a uniform positive gap. For example, the all-radix-one counter has orbit length $2\ell(L)$ and a closing gap of order $1/\ell(L)^2$, still with exponent zero.

## 7 Necessity from finite algebraic spectra

**Proposition 7.1.** *For fixed algebraic local matrices, every defined finite existing exponent is $\Delta_2^0$. If both endpoints are finite, the lower endpoint is in left-$\Sigma_2^0$ and the upper endpoint is in left-$\Pi_2^0$. This holds in any fixed lattice dimension and for any uniformly computable choice of finite boxes.*

*Proof.* For each $L$, construct $H_L$ exactly over its algebraic coefficient field. Exact characteristic polynomials and real-root isolation compute its distinct eigenvalues and decide multiplicities. Thus the gap above the entire ground space is a uniformly computable positive algebraic number whenever defined. The hypothesis that it is eventually defined is essential; the zero Hamiltonian has no second distinct eigenvalue. We do not replace algebraic entries by arbitrary computable real entries while retaining a multiplicity algorithm.

Compute rational $q_L$ within $1/L$ of $z_L$. This preserves all finite endpoints. For rational $r$,

$$r < \liminf_L q_L \quad \Longleftrightarrow \quad \exists t \in \mathbb{Q}\ \exists N \quad (r < t \text{ and } \forall L \ge N :\ q_L > t).$$

This is a $\Sigma_2^0$ lower cut. Dually the strict upper cut of $\limsup_L q_L$ is $\Sigma_2^0$. If that endpoint is irrational, its strict lower rational cut is the complement of its strict upper cut, hence $\Pi_2^0$; if it is rational, its lower cut is computable and is also $\Pi_2^0$. This avoids a false complement identity at a rational endpoint. An existing limit is computable in the halting oracle: that oracle can find arbitrarily accurate tail-Cauchy intervals of the computable sequence. Equivalently it is a $\Delta_2^0$ real.

For a $d$-dimensional box the norm estimate $\Delta(L) \le 2\|H_L\| \le CL^d$ gives an eventual lower bound for $z_L$. A finite limsup gives bounded tails when needed. No efficient diagonalization or computable rate of asymptotic convergence is claimed. □

This proves the remaining direction of Theorem 1.1. More generally, the same argument applies to total computable postprocessing of uniformly computable finite-scale spectral data. Any logarithm argument must be positive and every scale denominator must be nonzero. Merely naming a scale does not supply these domain conditions.

# 8 Physical scope and open problems

## 8.1 Frustration-free criteria require their hypotheses

Gosset and Mozgunov give local gap thresholds and, in their gapless one-dimensional periodic setting, an $O(n^{-2})$ open-chain upper bound [7]. Lemm and Mozgunov give finite-size criteria with boundaries [8]; their two-dimensional criterion uses a minimum over rhomboidal patches, and general boundaries require the corresponding intersected patch shapes. A square-box exponent alone does not provide those patch or edge estimates. Masaoka, Soejima and Watanabe obtain $z \ge 2$ under their frustration-free correlation hypotheses, including the extended correlations treated there [9]. These results do not imply an unconditional exclusion interval for the square-box exponents of the present setting.

The specified uniform local decomposition here is frustrated: an onsite flag reward competes with interior bond penalties. For a particular box, adding $4LI$ lets one regroup the flag energy into positive interior and boundary terms. That boundary-dependent regrouping does not identify the original interaction with a uniformly frustration-free family. No general necessity of frustration for every small exponent is proved here.

## 8.2 Changing to bulk or iterated-limit data

The upper approximation to a bulk gap in [12] concerns its operator-algebraic definition. It does not identify the open-box gap with the bulk gap or provide uniform momentum-resolved dispersion data. For clarity, even an abstract use of upper semicomputability must account for the sign of an inverse-gap normalization.

Suppose $g_n > 0$ is uniformly right-c.e., $c_n > 0$ is uniformly computable, and $x_n = -\log(g_n)/c_n$ is bounded. Then $x_n$ is uniformly *left*-c.e. Its liminf has a left-$\Sigma_3^0$ cut, its limsup has a left-$\Pi_2^0$ cut, and any existing limit is left-$\Pi_2^0$. Indeed for increasing rational approximants $x_{n,s}$, $r < \liminf_n x_n$ is expressed by $\exists t > r\ \exists N\ \forall n \ge N\ \exists s:\ x_{n,s} > t$. For the limsup, the non-strict cut has the $\Pi_2^0$ expression

$$r \le \limsup_n x_n \quad \Longleftrightarrow \quad \forall t \in \mathbb{Q}\ \forall N\ \exists n \ge N\ \exists s : \big(t \ge r \text{ or } x_{n,s} > t\big).$$

At an irrational endpoint it agrees with the strict lower rational cut; rational endpoints have computable cuts. An existing limit is its own limsup. These are arithmetic ceilings, not physical realizations.

The sign matters. Choose $\alpha \in$ left-$\Pi^0_2 \setminus$ left-$\Sigma^0_2$ in $[1,2]$ and a computable rational sequence $u_m \in [1,2]$ with limsup $\alpha$. Then $b_n = \sup_{m \geq n} u_m$ is uniformly left-c.e. and decreases to $\alpha$. The numbers $g_n = 2^{-nb_n}$ are positive uniformly right-c.e., even with known lower bounds $2^{-2n}$, but $-\log(g_n)/\log(2^n) = b_n \to \alpha$. Thus right-c.e. gaps alone do not force a left-$\Sigma^0_2$ existing inverse-gap exponent.

If a further limit is taken from uniformly $\Delta^0_2$ finite-scale data, the relativized necessity proof gives an existing $\Delta^0_3$ limit and left-$\Sigma^0_3$/left-$\Pi^0_3$ lower/upper endpoints. Such a statement requires uniform data and defined normalizations. It supplies no example of a physical exponent outside $\Delta^0_2$.

### 8.3 Questions left open

Can the full nonnegative classification be obtained for one-dimensional chains, for periodic two-dimensional boxes, or with a local dimension fixed independently of the prescribed pair? A one-dimensional realization would need a finite reversible compiler and a lower bound on every invalid sector; an orbit-length estimate alone is insufficient. No exponent range for chains is established here. Can an uncomputable exponent be realized in bulk dispersion, with a suitable uniform definition, or as a stable thermodynamic critical exponent? The boundary construction gives neither result. These questions require additional constructions and cannot be settled by counting the formal limits in a definition.

## How this work was done

The authors developed the argument with AI models from more than one developer and take responsibility for its mathematics, attribution and presentation. The proof uses a finite classical rule specification, graph-Laplacian estimates and computability arguments. Its claims concern all sectors, rather than a selected computational trajectory. The local matrices can be generated from the finite machine and rule specification; arbitrary endpoint programs do not come with a numerical many-body diagonalization certificate.

The ancillary directory contains a sequential runner, readable scripts, recorded outputs and SHA-256 checksums. The counter check constructs full finite register product spaces and numerically diagonalizes transition components, including illegal digits and all head numbers. The classical check minimizes over all counter/three-flag configurations on small boxes by a profile dynamic program. The penalty check tests bounded graphs. An independent direct-state implementation also checks the parallel-edge exception, frozen exterior sites, the four-flag low-energy enumeration, carry rules and exact rational word identities. The general estimates are proved in the text. Floating-point eigenvalues are finite consistency checks, not interval-certified proofs; neither the full two-dimensional tableau with an arbitrary program nor a noncomputable limit is numerically certified. The ancillary README specifies the coverage and reproduction commands.

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